



**Barcelona
Supercomputing
Center**

Centro Nacional de Supercomputación

BSC-CNS C6 Building Printer Manual

(v 1.0)

Barcelona, October 2024

Table of Contents

1	Introduction	3
1.1	Centralized Printing	3
1.2	C6 Building Printer Models and Printer IP addresses	3
2	Windows 10	4
2.1	Configuring the printer	4
3	Linux	11
3.1	Requirements.....	11
3.1.1	Ubuntu 22.04 (LTS).....	11
3.2	Configuring the printer	12
4	macOS.....	17
4.1	Requirements.....	17
4.2	Configuring the printer	17
5	History	22

1 Introduction

1.1 Centralized Printing

Currently the printers on C6 building are not integrated on BSC centralized print system.

1.2 C6 Building Printer Models and Printer IP addresses

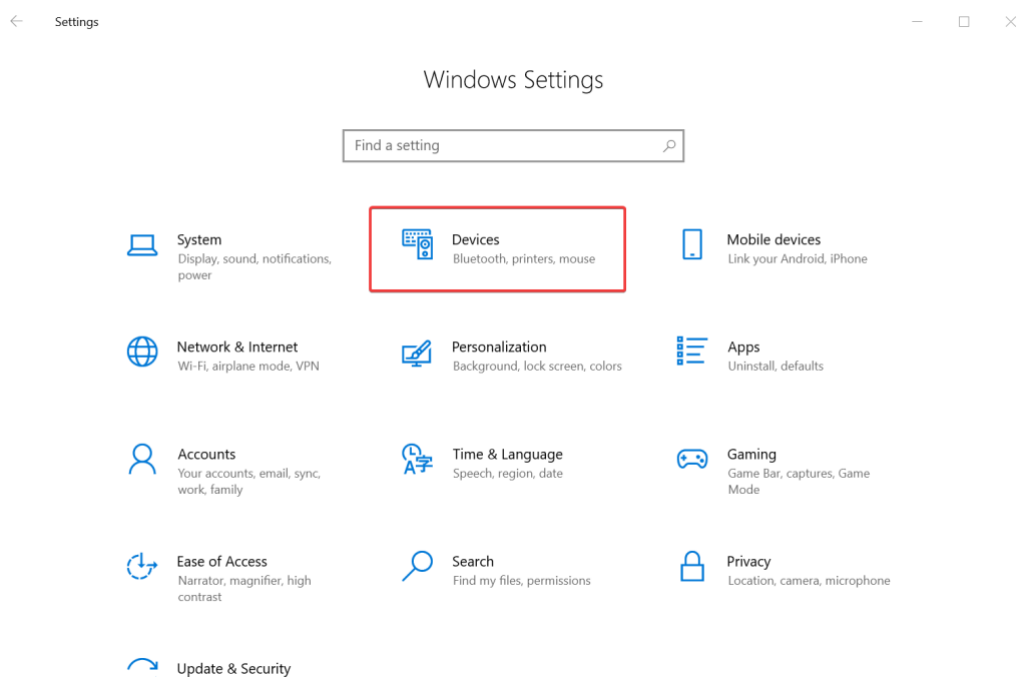
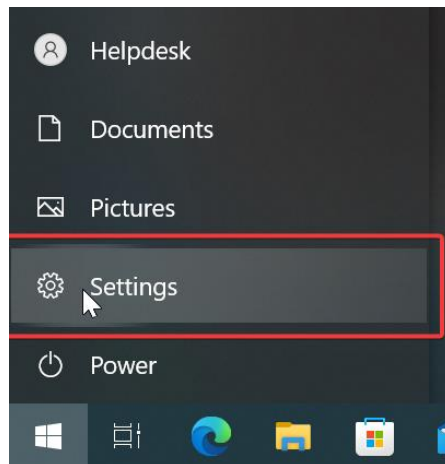
There are currently 2 printer models on C6 building, which requires different drivers on install. Bellow there is a table with each printer location, the model and the IP address required to install it.

Location	Printer Model	IP address
C6 – 002	Brother HL-L2360DN	192.168.3.208
C6 – E201	HP Laserjet P3015dn	192.168.3.202

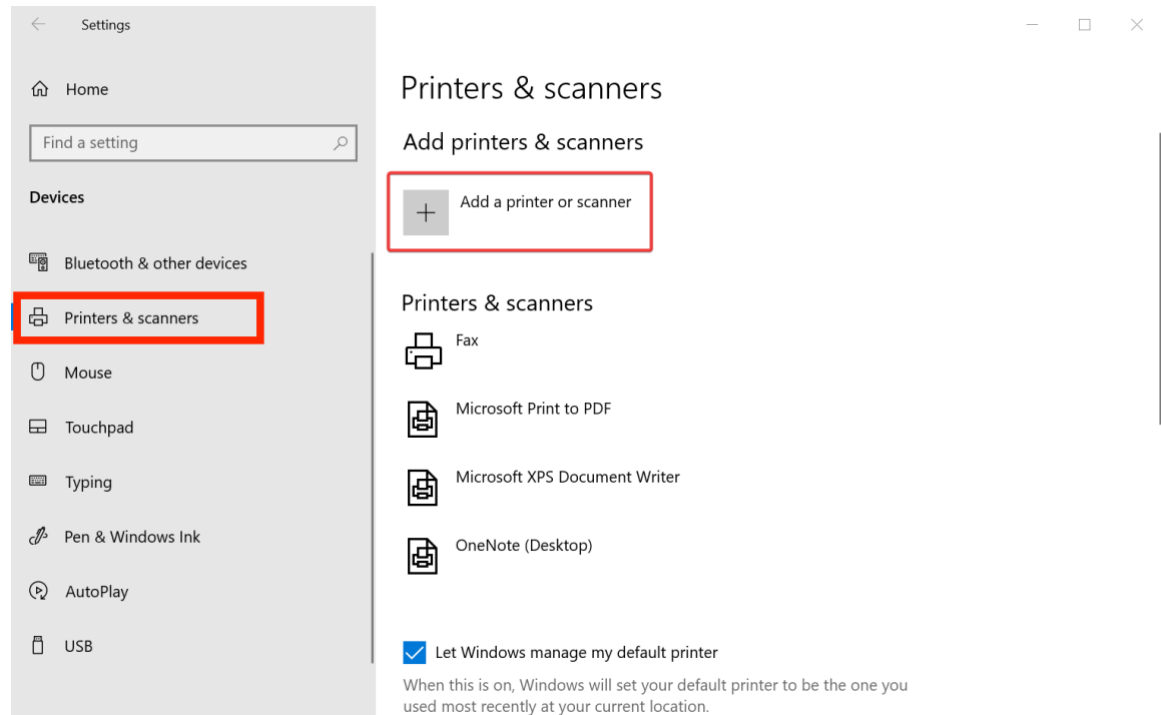
2 Windows 10

2.1 Configuring the printer

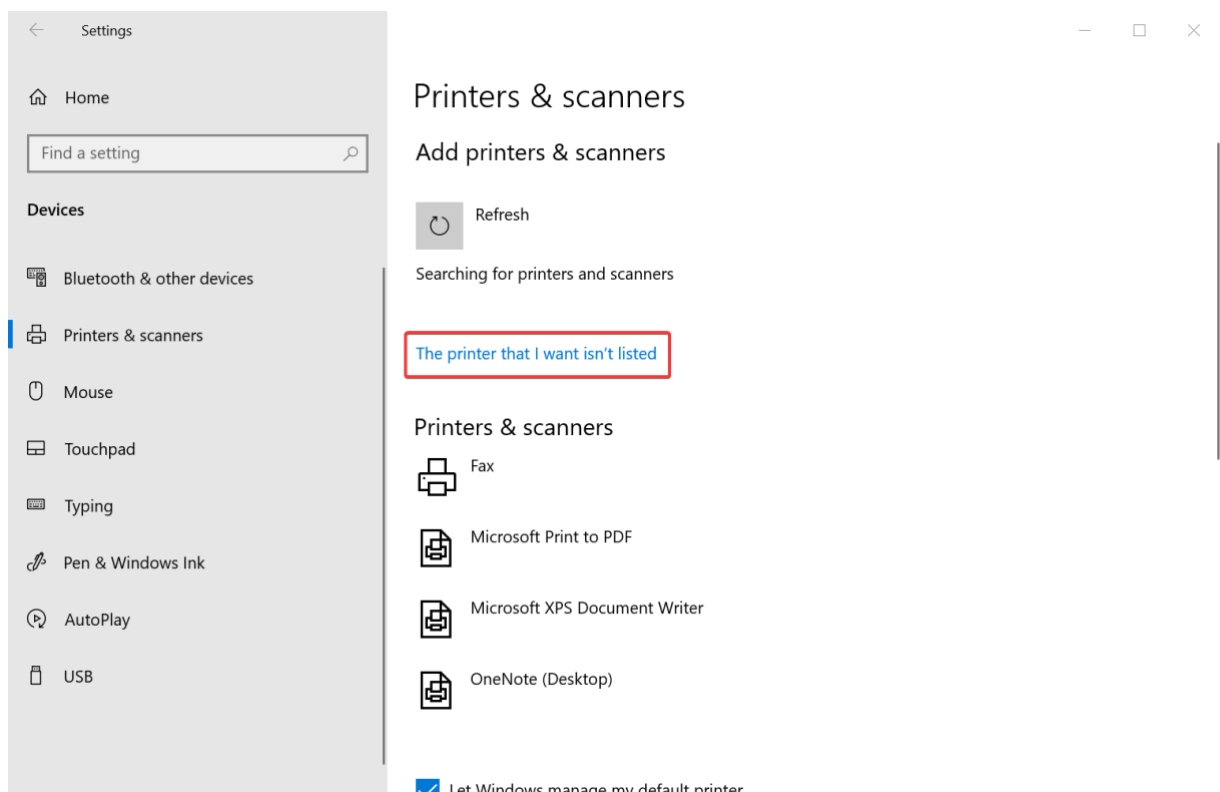
Open the Start menu, and click on *Settings*, and select *Devices*:



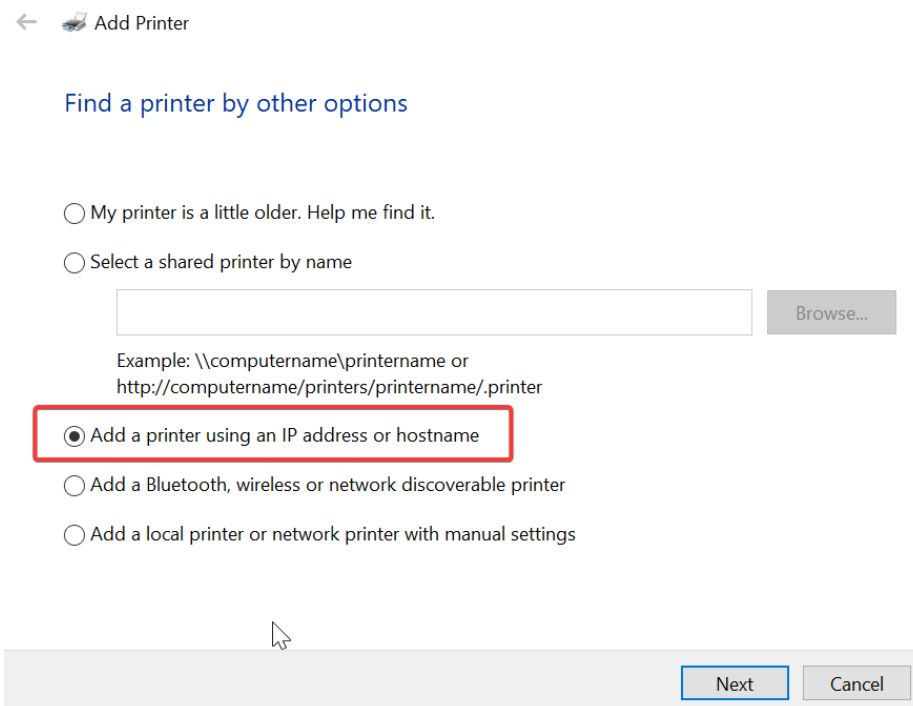
On the sidebar select “*Printers & scanners*” and press “*Add a printer or scanner*”



Wait while Windows search for a printer, it will fail to find the printer, after a while it will appear the option “*The printer that I want isn't listed*”.



On the next screen select “*Add a printer using an IP address or hostname*”:



← Add Printer

Find a printer by other options

☐ My printer is a little older. Help me find it.

☐ Select a shared printer by name

Browse...

Example: \\computename\prntername or
http://computename/printers/prntername/.printer

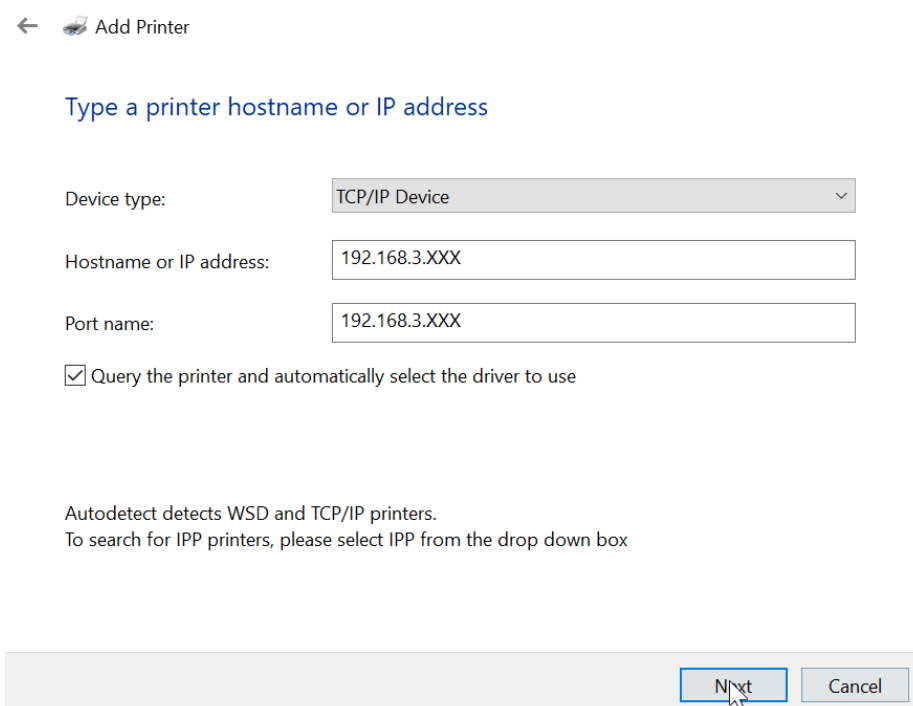
☒ Add a printer using an IP address or hostname

☐ Add a Bluetooth, wireless or network discoverable printer

☐ Add a local printer or network printer with manual settings

Next Cancel

Select “*Device Type*” to “*TCP/IP Device*” and enter the IP address of the printer you want to install, you can find the printers IPs on [1.2 C6 Building Printer Models and Printer IP](#) addresses



← Add Printer

Type a printer hostname or IP address

Device type: TCP/IP Device

Hostname or IP address: 192.168.3.XXX

Port name: 192.168.3.XXX

☒ Query the printer and automatically select the driver to use

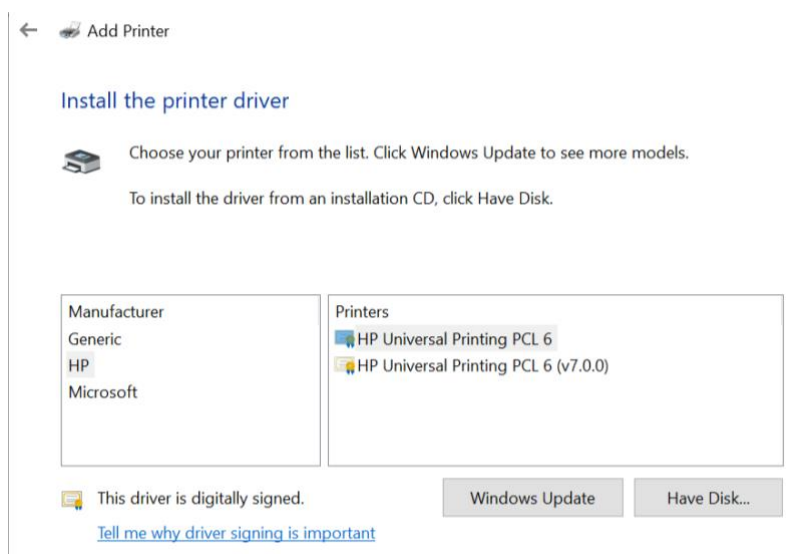
Autodetect detects WSD and TCP/IP printers.
To search for IPP printers, please select IPP from the drop down box

Next Cancel

Depending on the printer that you are installing, select one driver or the other:

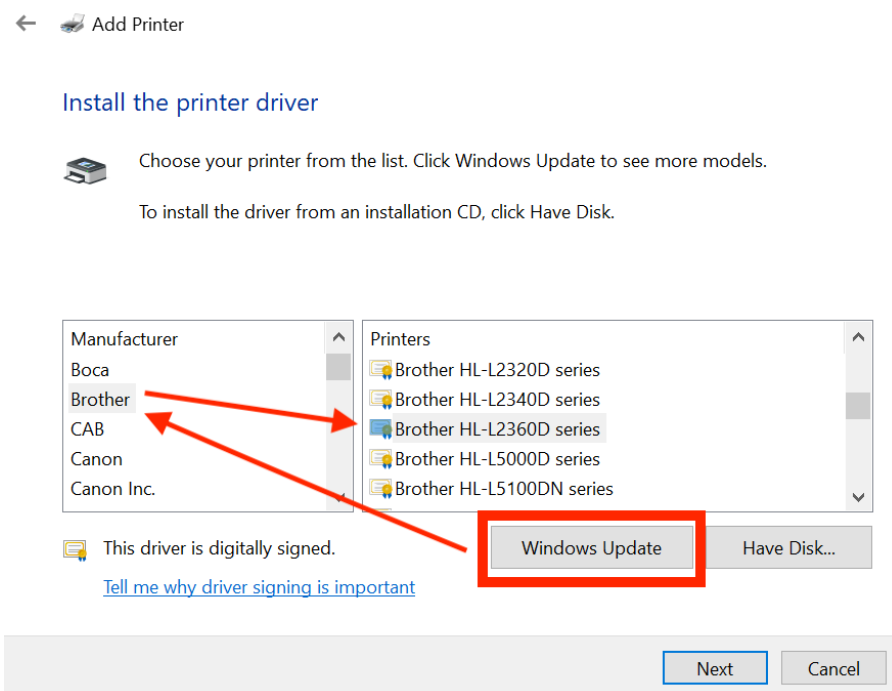
HP Printer Driver

Select “HP” as **Manufacturer** and “HP Universal Printing PCL 6” as the **printer**

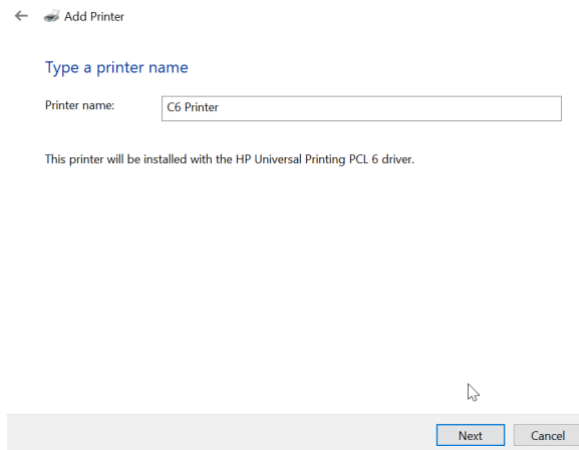


Brother Printer Driver

Click on “Windows Update” to download the driver for the Brother printer, after a while, the screen will be populated with more drivers, select “Brother” as **Manufacturer** and “Brother HL-L2360D series” as the **printer**.



After selecting the driver, enter a printer name, for example “C6 Printer”



← Add Printer

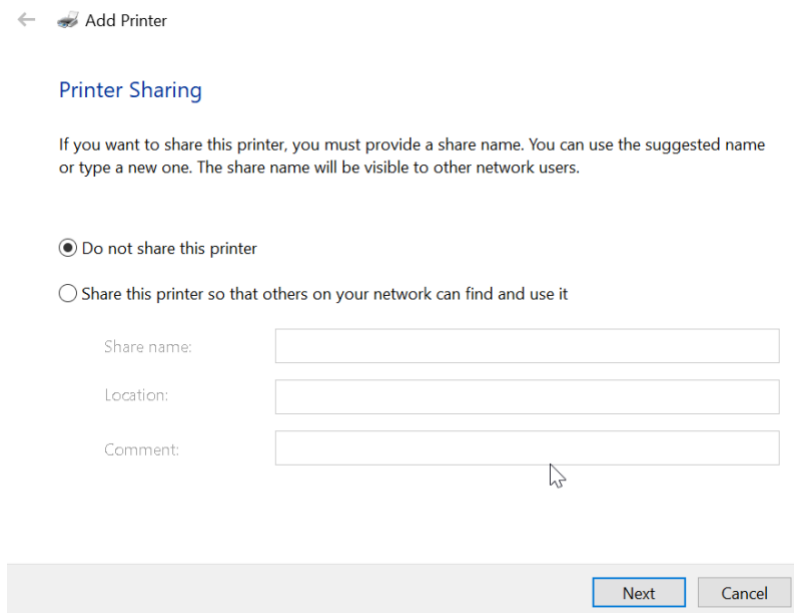
Type a printer name

Printer name:

This printer will be installed with the HP Universal Printing PCL 6 driver.

Next Cancel

And select “Do not share this printer”



← Add Printer

Printer Sharing

If you want to share this printer, you must provide a share name. You can use the suggested name or type a new one. The share name will be visible to other network users.

☒ Do not share this printer

☐ Share this printer so that others on your network can find and use it

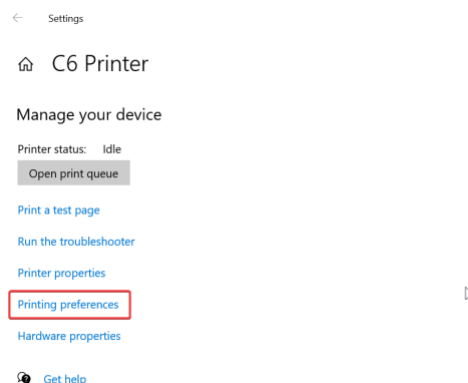
Share name:

Location:

Comment:

Next Cancel

After the printer is installed ensure, we need to configure it to ensure that it uses A4 and duplex options are enabled. Select the printer and click “*Printing preferences*”.



← Settings

C6 Printer

Manage your device

Printer status: Idle

Open print queue

Print a test page

Run the troubleshooter

Printer properties

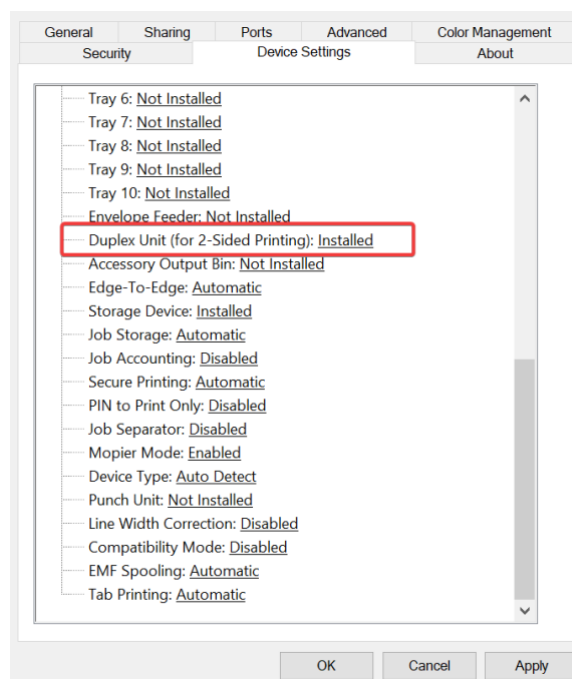
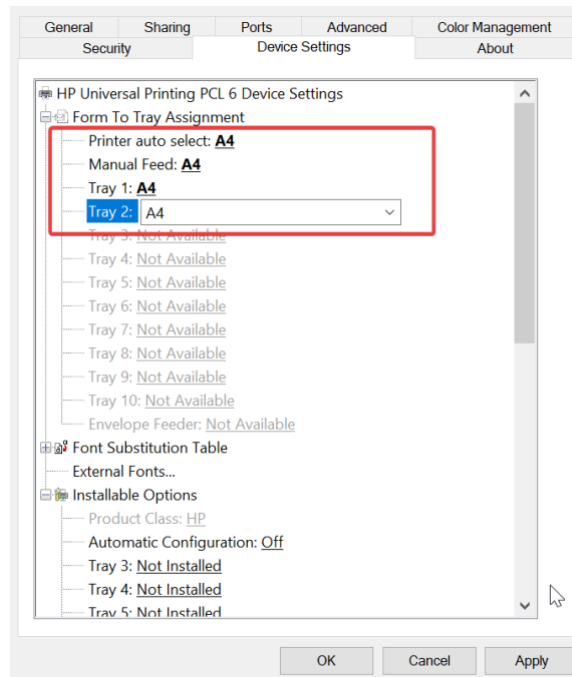
Printing preferences

Hardware properties

Get help

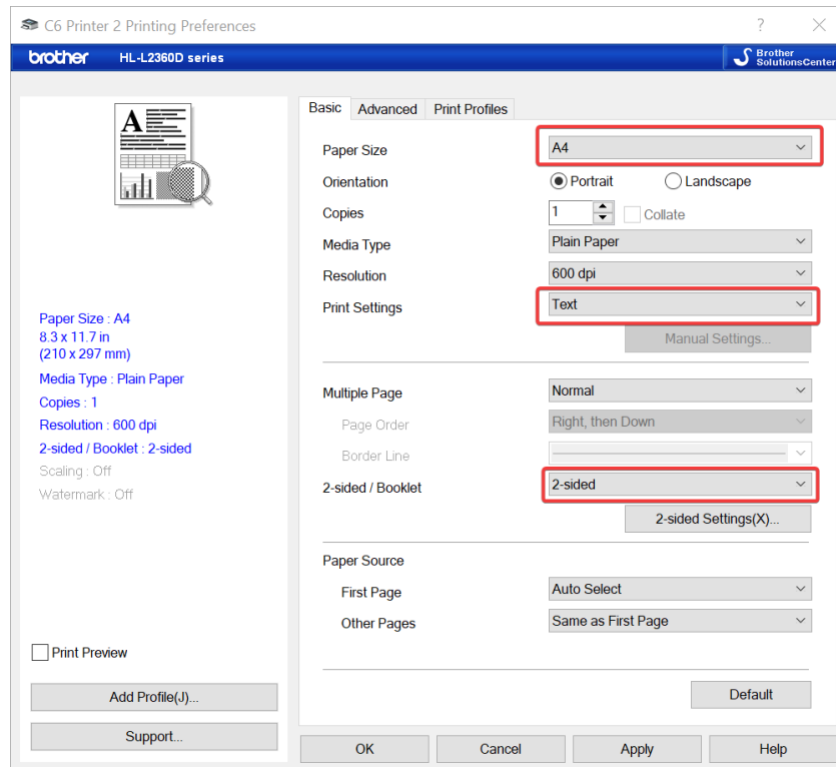
HP Printer preferences

Ensure that A4 is selected, and “*Duplex Unit*” is marked as **installed**



Brother Printer preferences

Ensure that A4 is selected, **Print Settings** is set to “Text” and “2-sided” is selected.



3 Linux

3.1 Requirements

To install the printer, you need the following packages installed on your Linux:

- HPLIP

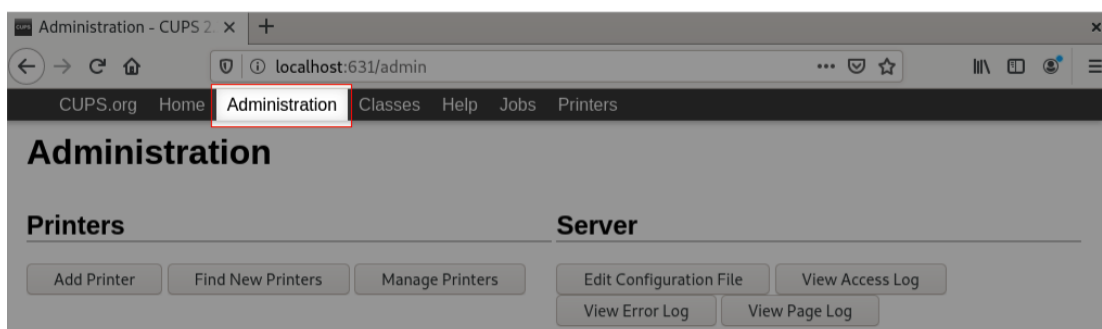
3.1.1 Ubuntu 22.04 (LTS)

On Ubuntu 22.04 you can install them running the following command:

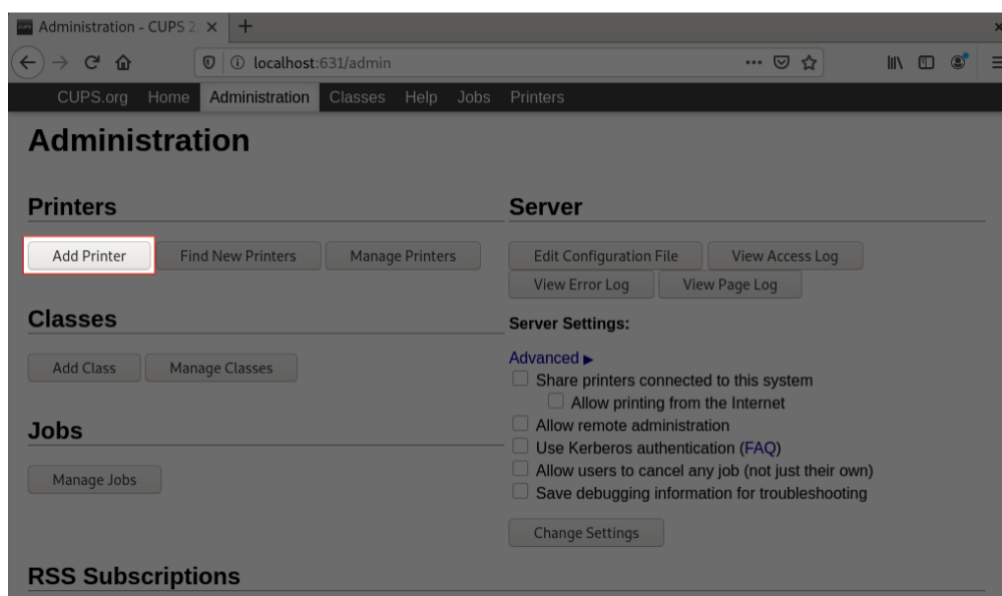
```
> sudo apt install hplip
```

3.2 Configuring the printer

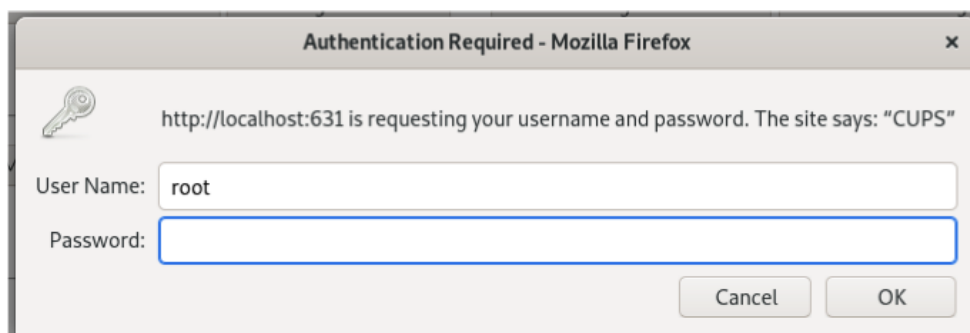
In a web browser enter this URL: <http://localhost:631> (use Firefox, as Chrome does not work correctly) and select the *Administration* tab:



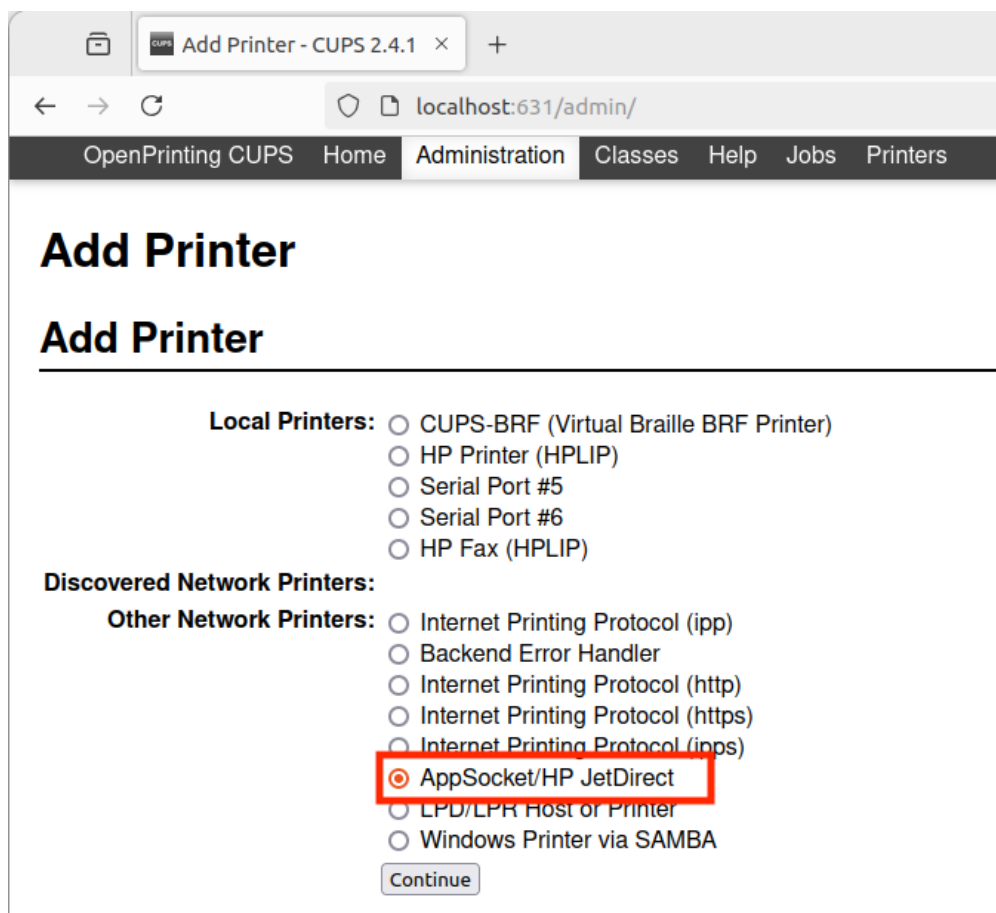
Select *Add Printer*



If the browser, ask for a username and a password write your local administrator account (on Linux usually root). This is not your BSC account.



Select the option “AppSocket/HP JetDirect”:

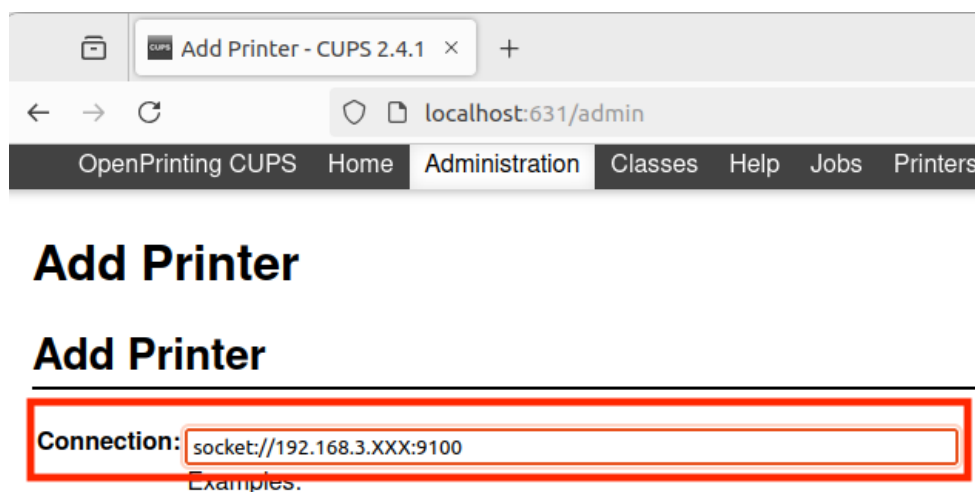


The screenshot shows the 'Add Printer' page in the OpenPrinting CUPS administration interface. The browser address bar shows 'localhost:631/admin/'. The page has a navigation bar with 'OpenPrinting CUPS', 'Home', 'Administration', 'Classes', 'Help', 'Jobs', and 'Printers'. The main heading is 'Add Printer'. Below it, there are two sections: 'Local Printers' and 'Discovered Network Printers'. Under 'Local Printers', there are radio buttons for 'CUPS-BRF (Virtual Braille BRF Printer)', 'HP Printer (HPLIP)', 'Serial Port #5', 'Serial Port #6', and 'HP Fax (HPLIP)'. Under 'Discovered Network Printers', there is a sub-section 'Other Network Printers' with radio buttons for 'Internet Printing Protocol (ipp)', 'Backend Error Handler', 'Internet Printing Protocol (http)', 'Internet Printing Protocol (https)', 'Internet Printing Protocol (ipps)', 'AppSocket/HP JetDirect' (which is selected and highlighted with a red box), 'LPD/LPR Host or Printer', and 'Windows Printer via SAMBA'. A 'Continue' button is at the bottom.

On *Connection* write the IP for the printer as following:

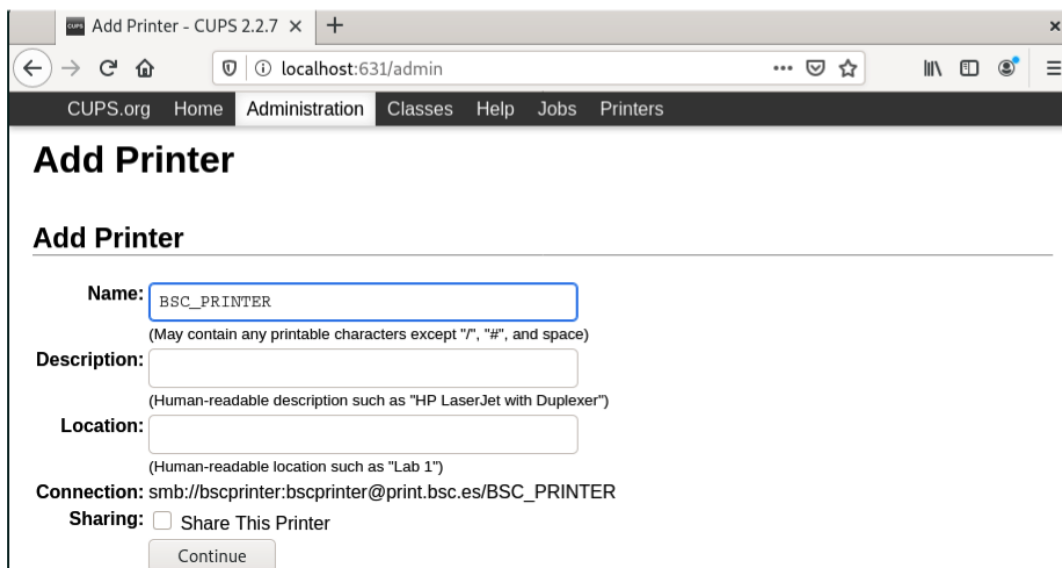
socket://192.168.3.XXX:9100

Change XXX for the correct IP address of the printer. You can find the printers IPs on [1.2 C6 Building Printer Models and Printer IP](#) addresses



This screenshot shows the same 'Add Printer' page, but now the 'Connection' field is highlighted with a red box. The field contains the text 'socket://192.168.3.XXX:9100'. Below the field, the word 'Examples.' is partially visible.

Write the name that you want to use to identify the printer on your computer.



Add Printer

Name: BSC_PRINTER
(May contain any printable characters except "I", "#", and space)

Description:
(Human-readable description such as "HP LaserJet with Duplexer")

Location:
(Human-readable location such as "Lab 1")

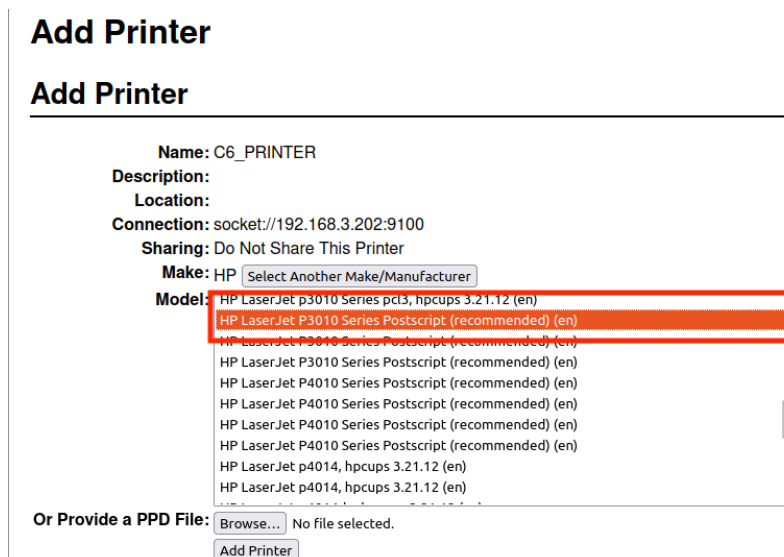
Connection: smb://bscprinter:bscprinter@print.bsc.es/BSC_PRINTER

Sharing: ☐ Share This Printer

We need to choose the driver, depending on the printer that you are installing, select one driver or the other:

HP Printer Driver

Select "HP" as the **Maker** and "HP LaserJet P3010 Series Postscript" as the **Model**.



Add Printer

Name: C6_PRINTER

Description:

Location:

Connection: socket://192.168.3.202:9100

Sharing: Do Not Share This Printer

Make: HP

Model: HP LaserJet p3010 Series pcl3, hpcups 3.21.12 (en)
HP LaserJet P3010 Series Postscript (recommended) (en)
 HP LaserJet P3010 Series Postscript (recommended) (en)
 HP LaserJet P3010 Series Postscript (recommended) (en)
 HP LaserJet P4010 Series Postscript (recommended) (en)
 HP LaserJet P4010 Series Postscript (recommended) (en)
 HP LaserJet P4010 Series Postscript (recommended) (en)
 HP LaserJet P4010 Series Postscript (recommended) (en)
 HP LaserJet p4014, hpcups 3.21.12 (en)
 HP LaserJet p4014, hpcups 3.21.12 (en)

Or Provide a PPD File: No file selected.

To finish click "Add Printer".

Now we check the “*Default Options*”. The options to check are:

Options Installed Tab:

Duplex Unit: Installed

Set Printer Options

Set Default Options for C6_PRINTER

Query Printer for Default Options

Options Installed Secure Printing Services Finishing Panel General Image Quality
Banners Policies Port Monitor

Options Installed

Collation in Printer: Yes: Total RAM >= 128 MB or Hard Disk ▾

Tray 3: Not Installed ▾

Tray 4: Not Installed ▾

Duplex Unit: ☒ Installed ☐ Not Installed

Printer Disk: None ▾

Set Default Options

General Tab:

Page Size: A4

2-Sided: “Long-Edge Binding”

Set Printer Options

Set Default Options for C6_PRINTER

Query Printer for Default Options

Options Installed Secure Printing Services Finishing Panel **General** Image Quality
Banners Policies Port Monitor

General

2-Sided Printing: Long-Edge Binding ▾

Collate: ☒ On (turn off in application) ☐ Off

Page Size: A4 ▾

Paper Source: Automatic ▾

Set Default Options

To finish click “*Set Default Options*”

Brother Printer Driver

Select “*Brother*” as the **Maker** and “*Brother HL-L2360D series using brlaser v6 (en)*” as the **Model**.

Add Printer

Name: C6_PRINTER_2
Description:
Location:
Connection: socket://192.168.3.208:9100
Sharing: Do Not Share This Printer
Make: Brother [Select Another Make/Manufacturer](#)
Model:

- Brother HL-8050N BR-Script3 (en)
- Brother HL-8050N Foomatic/Postscript (en)
- Brother HL-L2300D series, using brlaser v6 (en)
- Brother HL-L2320D series, using brlaser v6 (en)
- Brother HL-L2340D series, using brlaser v6 (en)
- Brother HL-L2360D series, using brlaser v6 (en)**
- Brother MC-3000 Foomatic/epson (recommended) (en)
- Brother MFC 7150C Foomatic/epsonc (recommended) (en)
- Brother MFC-1910W, using brlaser v6 (en)
- Brother MFC-6550MC Foomatic/ljet2p (recommended) (en)

Or Provide a PPD File: [Browse...](#) No file selected.
[Add Printer](#)

To finish click “*Add Printer*”.

Now we check the “*Default Options*”. The options to check are:

General Tab:

Media Size: A4

2-Sided Printing: “Long-Edge (Portrait)”

Set Printer Options

Set Default Options for C6_PRINTER_2

General **Banners** **Policies**

General

Media Size: A4
Resolution: 600 DPI
Media Source: Auto-select
Media Type: Plain paper
Toner save mode: ☒ Off ☐ On
2-Sided Printing: Long-Edge (Portrait)

[Set Default Options](#)

To finish click “*Set Default Options*”

4 macOS

4.1 Requirements

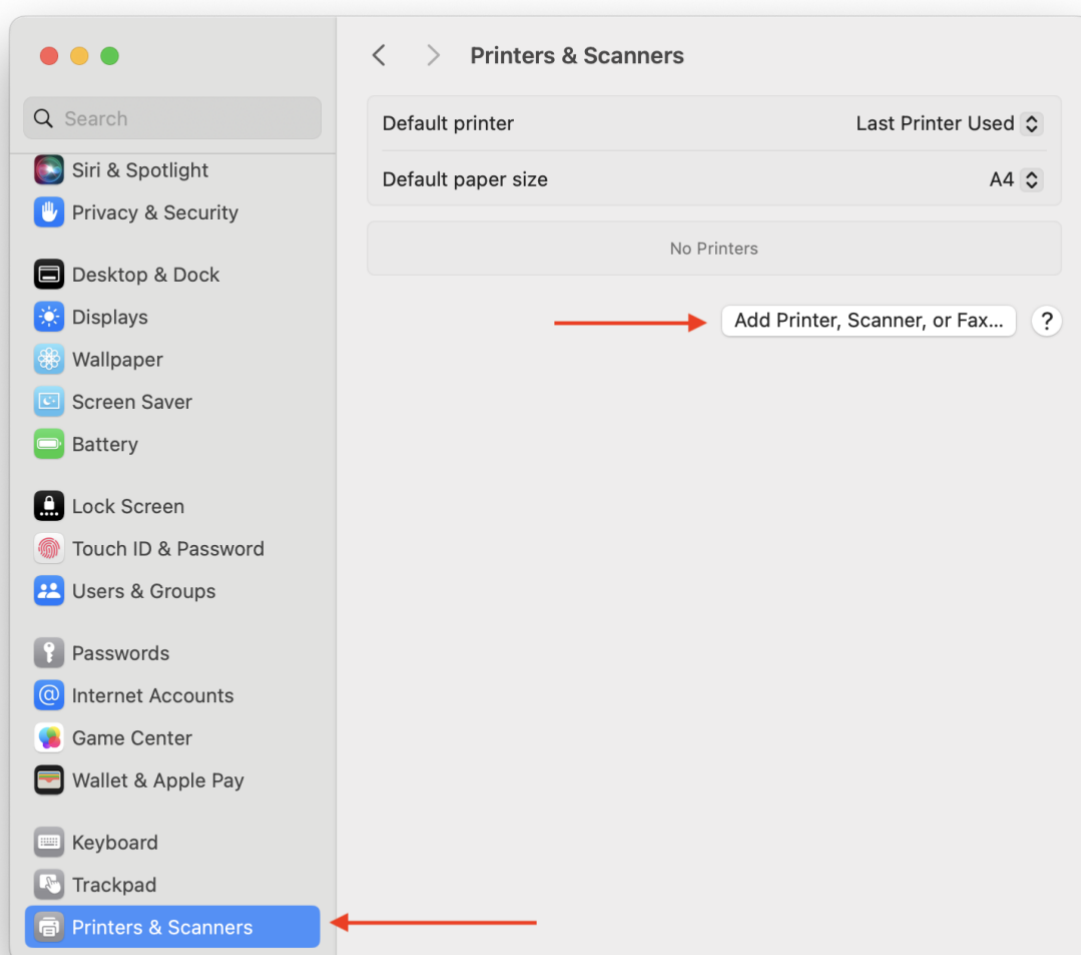
If you want to use the Brother printer, we need to install the printer drivers that can be found at the BSC Nextcloud service, on the folder, BSC-CNS/Helpdesk/Printers/macOS:

<https://nextcloud.bsc.es/index.php/apps/files/?dir=/BSC-CNS/Helpdesk/Printers/macOS>,

the filename is “*Brother_PrinterDrivers_MonochromeLaser_1_2_0.dmg*”.

4.2 Configuring the printer

Choose “*Apple menu > System Preferences*”. Then click “*Printers & Scanners*”:



On the “Printers & Scanners” window click on the “*Add Printer, Scanner, or Fax...*” button to set up the printer.

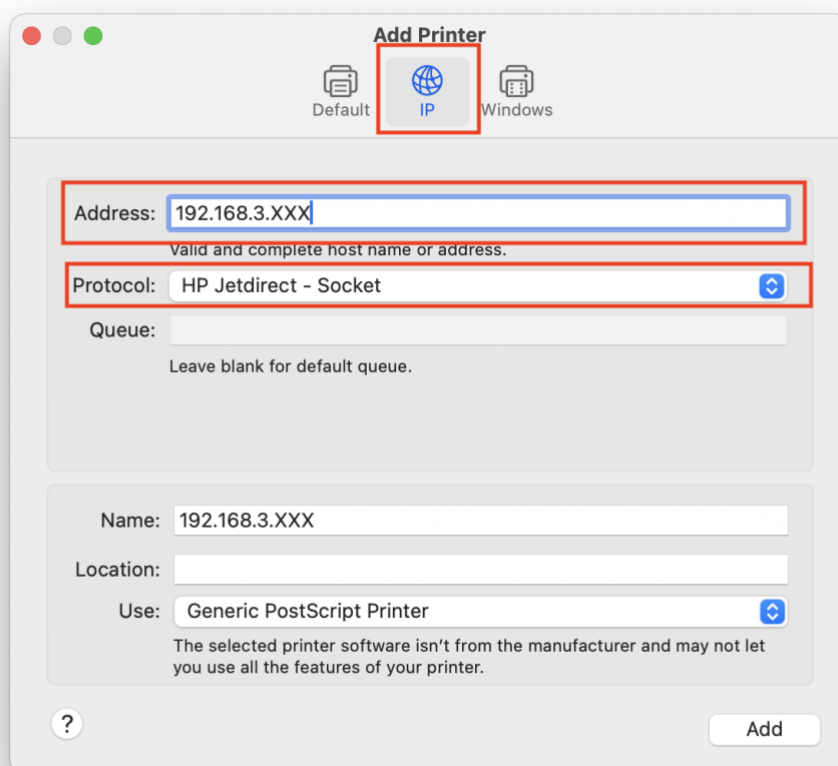
On the “Add Printer” window, select the “IP” tab, to configure the printer.

On **Address** enter the IP address of the printer you want to install, you can find the printers IPs on [1.2 C6 Building Printer Models and Printer IP addresses](#)

Select “*HP Jetdirect – Socket*” as **Protocol**

And on **Name**, write a name you want, for example “*C6 Printer*”

On **Use**, select the driver depending on the printer that you are installing

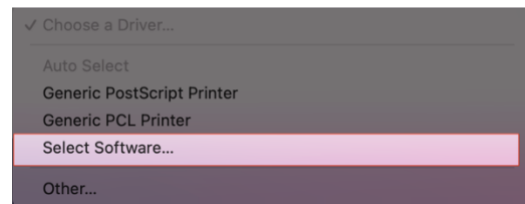


HP Printer Driver

Use the default “*Generic PostScript Printer*”

Brother Printer Driver

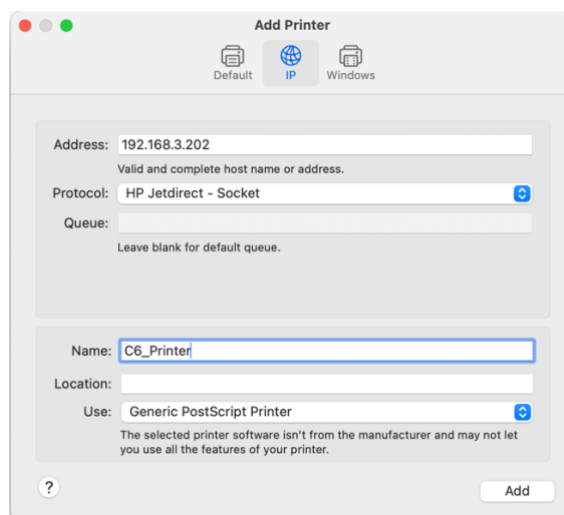
On the “Choose a Driver...” option select: “Select Software...”:



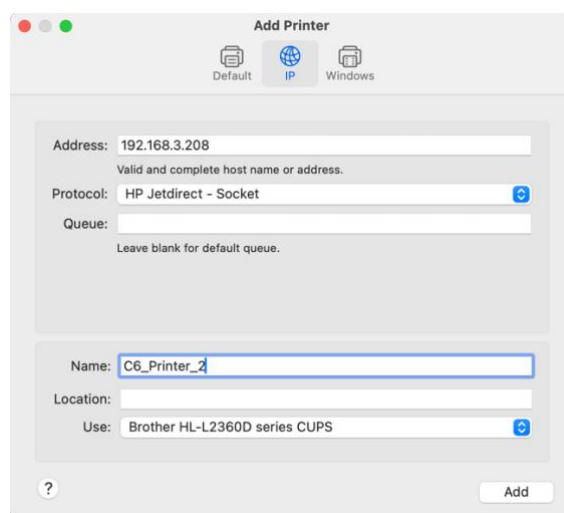
Select the driver “*Brother HL-L2360D series CUPS*”

Now we can add the printer:

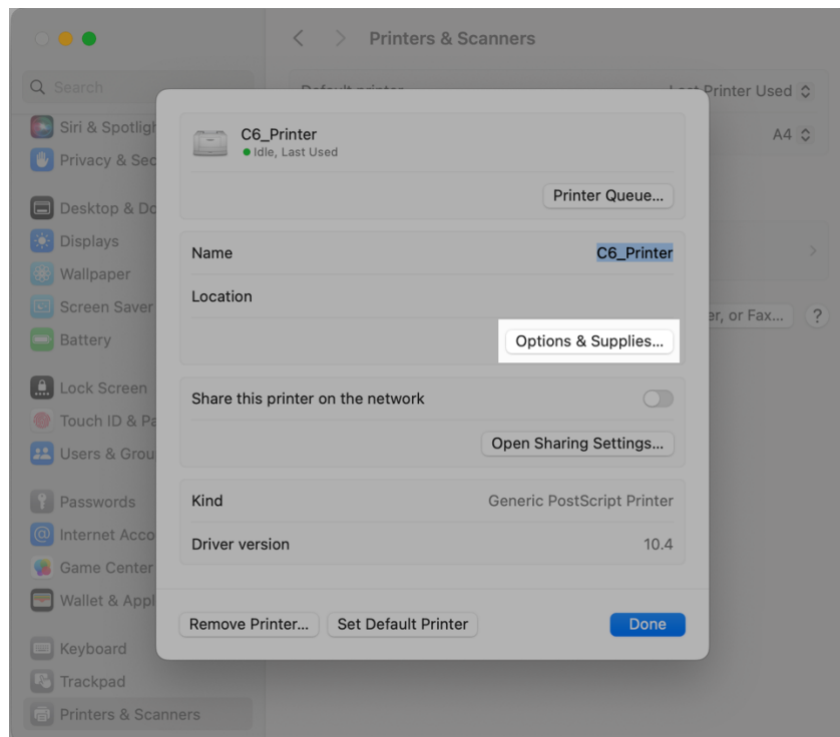
HP



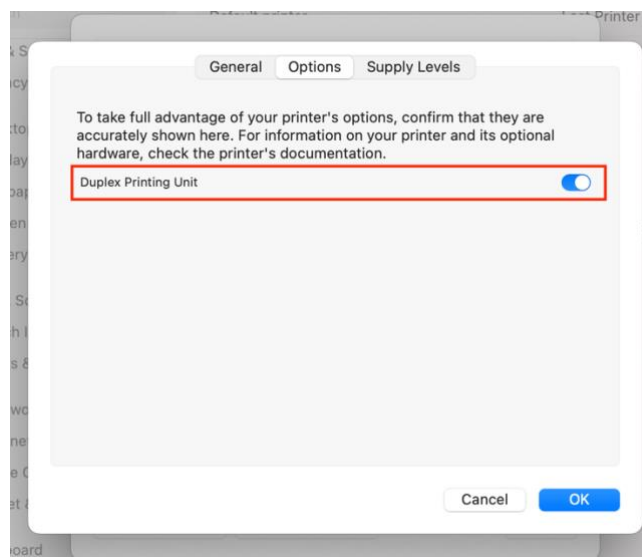
Brother



After installing the HP printer go to “Options & Supplies”:



And ensure that in “Options” the “Duplex Printing Unit” is enabled:



5 History

1.0 09/10/2024

Initial release